

TAYLKOMB 2026 Size, Ergonomics, Weight, and Geometry Finalization for CAD Handoff

Executive verdict

TAYLKOMB's draft lengths (main body ~208 mm; connectors ~243 mm; attachments ~223 mm) sit broadly within **professional comb length bands** seen in premium benchmarks: shorter "cutting" combs around **180 mm**, tail combs around **216 mm**, and long detangling/rake combs around **~229 mm**. ¹ The core opportunity is not "reinvent length," but to **standardize handling** (overall length families), **reduce snag risk** (seam strategy + tooth/edge finishing), **make grip geometry match real hand biomechanics**, and **choose a modular interface that survives 10,000+ swaps without wobble**.

Biggest lock-now vs. prototype-first guidance

Lock-now (good confidence; benchmark-backed + engineering rules):

- **Overall length families:**
 - "Cutting/precision" family target **185–205 mm** (shorter, faster in-hand). ²
 - "Tail/sectioning + long styling" family target **215–235 mm** (industry-common). ³
 - "Detangling/rake/curly" family target **225–240 mm**. ⁴
- **Grip sizing rules** (hand biomechanics):
 - Precision-grip effective diameter **8–16 mm**, nominal **~12 mm**. ⁵
 - Power-grip effective diameter **30–50 mm** (often too large for combs, but a target for "double-handle" or palm-supported portions). ⁵
- **DFM radii:** feature radii should generally be **25–40% of local wall thickness**, with a **minimum absolute radius ~0.5 mm** in demanding plastic parts. ⁶
- **Snap-fit strain safety** (if any snap is used): favor **tapered cantilever arms** (reduces strain concentration) and design permissible strain conservatively (e.g., amorphous up to ~70% of yield strain; GF materials ~½ elongation at break), with flush, burr-free surfaces. ⁷

Prototype-first (must validate physically):

- Exact **tooth pitch, tooth length, and tooth-root thickness** for wide/narrow attachments (controls snagging vs. stiffness vs. comfort). Benchmarks rarely publish these, so this is intrinsically prototype-driven. ⁸
- **Connector wobble vs. insertion force trade** (tolerances, wear surfaces, hair/debris contamination). Injection molding tolerances and resin shrink variability mean "perfect on CAD" can still wobble in production if the interface is too short or too lightly constrained. ⁹
- **Weight and balance point:** for an all-day stylist tool, small weight shifts can change perceived control; validate with weighted prototypes (simple ballast test) despite theoretical targets. ¹⁰

High-level redesign directive for CAD (next revision)

- 1) **Rationalize system into two consistent handling “families”** (short precision vs long detangle/section) rather than mixing lengths across modules unpredictably. ¹
- 2) Treat “premium feel” as: **seamless glide, no snag edges, generous radii in hand-contact zones, no sharp transitions, and repeatable modular lock geometry**. This matches why premium combs emphasize “no parting lines,” symmetry, and smooth glide. ¹¹
- 3) Pick a modular interface that is **anti-rotation + anti-wobble by geometry (keyed rails/polygon), retention by detent, and protected from hair ingress** (flush seam + hair-shed chamfers). Snap fits can work, but must be designed to correct strain and wear principles. ¹²
- 4) Decide your disinfection/chemical-exposure stance: some premium comb brands explicitly advise **soap and water only**—that conflicts with typical salon immersion disinfectants. Your material and surface decisions must reflect your target salon protocol. ¹³

Part-by-part audit

Legend used throughout (per your guardrails):

- **Benchmark-backed**: directly supported by comparable pro tools or published specs.
- **Engineering inference**: derived from ergonomics/DFM principles applied to your geometry.
- **Requires prototype validation**: cannot be responsibly locked without physical testing.

Main comb body

Current draft dimensions (from your prompt):

- Overall body length: **~208.01 mm** (other measurements: **not found in chats**).

What likely works

- **208 mm** is within a practical “general comb” length and near the common pro range for longer cutting/section combs, supporting control without being unwieldy. (Benchmark-backed) ³
- If this body is the “spine” that defines handling, keeping it near ~200–220 mm can maintain muscle memory across attachments. (Engineering inference) ⁸

What likely does not work / risk

- If the modular seam or connector socket sits near the tooth zone, you risk **hair snagging** unless the seam is shielded and flush (premium combs emphasize eliminating parting lines / snag points). (Benchmark-backed → performance expectation) ¹¹
- If body thickness is too thin (<~2 mm in structural zones), stiffness and long-term durability drop; if too thick (>~4 mm) you risk sink/warp and heavy feel (DFM). (Engineering inference; wall-thickness targets below are informed by thin-wall molding guidance) ¹⁴

Recommended revised targets (next CAD)

- **Overall length (assembled “comb-body-defined” length)**
- Ideal: **210 mm** (Engineering inference)

- Acceptable: **200–220 mm** (Benchmark-backed by comparison range) ³
- Too small: **<190 mm** (starts to feel like a short cutting comb; may conflict with modularity) (Benchmark-backed) ¹⁵
- Too large: **>230 mm** (begins to compete with rake/tail lengths; can feel slow for precision) (Benchmark-backed) ⁴
- **Structural wall thickness (non-tooth structural spine zones)**
 - Ideal: **2.2–2.8 mm nominal**, with local ribs rather than thick sections (Engineering inference; supported by injection molding “uniform wall” guidance) ¹⁶
 - Too thin: **<1.8 mm** (risk: warp, flex, weak interface, sink sensitivity if compensated) (Inference) ¹⁷
 - Too thick: **>3.5–4.0 mm** in broad areas (risk: sink/warp, heavier feel) (Inference) ¹⁷
- **Hand-contact edge radii**
 - Apply feature radii broadly; target $R \approx 0.25\text{--}0.40 \times \text{wall thickness}$, and avoid **<0.5 mm** where structural integrity/manufacturing quality matters. (Benchmark-backed DFM) ⁶

Confidence level: Medium-high on overall length band; medium on thickness/radii pending chosen resin and molding approach.

Round handle connector

Current draft dimensions (from your prompt):

- Overall length: **~242.65 mm**
- Tip (diameter or thickness, assumed): **~3.11 mm**
- Other dims: **not found in chats.**

What likely works

- A long handle/connector in the **~230 mm class** is consistent with long styling/rake combs and is not “out of market.” (Benchmark-backed) ⁴
- If this component functions like a tail/sectioning handle, length near the pro tail comb region (**~216 mm**) is common; your draft is longer but not absurd. (Benchmark-backed) ¹⁸

What likely does not work / risk

- A **3.11 mm tip** is likely **thick** for clean parting lines if the goal is precise sectioning; many tail combs rely on a very fine tail (often metal) for precision, while plastic tails are necessarily thicker—this needs validation for your target technique. (Engineering inference; requires prototype validation for stylist preference) ¹⁸
- A perfectly round handle can reduce indexed orientation (you lose “which side is up” feel), unless you add a keyed flat or tactile index. (Engineering inference)

Recommended revised targets

- **Overall length** (choose based on intended role):
 - If “tail/sectioning + long control”:
 - Ideal: **225–235 mm** (Benchmark-backed by long combs; engineering inference to pull slightly shorter than 243 for speed) ⁸
 - Acceptable: **215–242 mm** (Benchmark-backed; aligns with 216 mm tail combs and ~229 mm rake combs) ¹⁹
 - Too short: **<210 mm** (starts to lose tail reach) (Inference)
 - Too long: **>245 mm** (fatigue/awkwardness risk for smaller hands, tool belt storage) (Inference)
- **Tip diameter/thickness** (precision parting):
 - Ideal: **2.0–2.6 mm** with a soft radius at tip (Engineering inference)
 - Acceptable: **1.8–3.0 mm** (Prototype validation; depends on scalp comfort vs line precision)
 - Too small: **<1.6 mm** (risk: bending, breakage, sharp-feel unless radius controlled) (Inference)
 - Too large: **>3.2 mm** (risk: coarse parting, less precision) (Inference)
- **Grip “effective diameter”**
 - Ideal for precision control: **10–14 mm** (benchmark-backed ergonomic range) ⁵
 - If you want dual-mode (precision + some power): consider ovalizing to **major 16–20 mm / minor 10–14 mm** (Engineering inference referencing precision range) ⁵

Confidence level: Medium (length), low-medium (tip diameter until stylist testing).

Flat handle connector

Current draft dimensions (from your prompt):

- “Long-body connector in similar dimensional family” → **unspecified** exact values.

What likely works

- Flat/oval handles provide orientation indexing and can support both pinch and light power grips better than truly round sections. (Engineering inference; consistent with handle ergonomics guidance about grip geometry) ²⁰

What likely does not work / risk

- Too thin (<~6–7 mm) increases pinch force and pressure hotspots; too wide without thickness can feel “flimsy” and twist in hand. (Engineering inference; validate via prototypes) ⁵

Recommended revised targets

- **Overall length:** align to your system family
- Ideal: **220–235 mm** (tail/long family) (Benchmark-backed band) ⁸
- Acceptable: **215–242 mm**
- **Cross-section (grip zone)**
- Ideal: **width 18–26 mm, thickness 8–12 mm**, with generous edge radii (Engineering inference; thickness aligns to precision grip diameter band) ²¹
- Too thin: **<7 mm** (pinch fatigue risk) (Inference) ⁵
- Too thick: **>14–16 mm** (starts to become “bulky”; may slow dexterity) (Inference)

Confidence level: Medium (grip size bands), requires quick foam/print validation.

Double handle connector

Current draft dimensions (from your prompt):

- Overall length: **~242.8 mm**
- Outer grip width: **~27.2 mm**
- Other dims: **not found in chats.**

What likely works

- A ~27 mm grip width is close to the **lower end** of commonly favored cylindrical/oval handle sizes used for “power-type” tasks, and it’s large relative to pencil-like precision handles—this can reduce pinch stress if shaped correctly. (Engineering inference informed by ergonomic diameter bands) ⁵
- Length ~243 mm is within the long comb class; feasible for detangling/long hair control if not too head-heavy. (Benchmark-backed band) ⁴

What likely does not work / risk

- If “double handle” implies two parallel members, hair may snag between them unless the geometry is shielded. (Engineering inference)
- If the interface connection is not anti-rotation keyed, wide grips can amplify perceived wobble at the comb end. (Engineering inference; tolerance/wear risk) ²²

Recommended revised targets

- **Overall length:**
- Ideal: **230–240 mm** (Benchmark-backed band; slightly shorter than 243 if you want faster handling) ²³
- Too long: **>245 mm** (Inference)
- **Grip cross-section target** (translate diameter guidance into an oval/rectangular “effective grip”):

- If intended for “power grip moments” (detangle/smooth):
 - Ideal effective diameter: **25–35 mm**; practical comb-friendly oval: **major 26–32 mm, minor 16–22 mm** (Engineering inference; anchored to power-grip band but scaled for a salon comb) ⁵
- Your current **27.2 mm outer width** is inside this recommended major-axis band—good starting point. (Inference) ⁵

Confidence level: Medium; needs prototype to validate snag risk and one-handed control.

Wide comb attachment

Current draft dimensions (from your prompt):

- Overall length: **~223.23 mm**
- Tooth geometry: **not found in chats**.

What likely works

- **~223 mm** matches the long comb class seen in premium long cutting combs (~8.75" ≈ 222 mm). (Benchmark-backed) ²⁴
- If this is meant for detangling/smoothing, a longer tool aligns with market and allows longer stroke length. (Benchmark-backed) ⁴

What likely does not work / risk

- Wide-tooth attachments are the most sensitive to: **tooth root cracking, tooth tip discomfort**, and **hair snag at the tooth base** (mold parting, sharp corners). Premium combs emphasize burr-free, snag-free, seamless feel for a reason. (Benchmark-backed expectation) ²⁵

Recommended revised targets

- **Overall length**
 - Ideal: **225–235 mm** (Benchmark-backed band; keep near current 223 but standardize) ⁸
 - Acceptable: **220–240 mm** (Benchmark-backed) ⁴
- **Tooth geometry (prototype-validation driven)**
 - Tooth length (from spine to tip): **35–55 mm** (Inference; validate on curly/thick hair)
 - Tooth pitch (center-to-center): **4–8 mm** (Inference; wide detangle class)
 - Tooth root thickness: **2.2–3.2 mm** (Inference; strength + molding)
 - Tooth tip thickness: **1.2–1.8 mm with rounded tip** (Inference; comfort)
 - Root fillet radius at each tooth: **R 0.6–1.2 mm** (Inference; based on radii guidance and local thickness) ⁶

Confidence level: High on overall length; low-medium on tooth dimensions until stylist trials.

Narrow comb attachment

Current draft dimensions (from your prompt):

- Overall length: ~**223.67 mm**
- Tooth geometry: **not found in chats**.

What likely works

- Length aligns with long comb benchmarks (~222 mm). (Benchmark-backed) ²⁴

What likely does not work / risk

- If this attachment is for cutting/precision, 223–224 mm may be “long” compared to the 180–191 mm range typical of cutting combs. That doesn’t make it wrong, but it changes speed and dexterity. (Benchmark-backed) ²

Recommended revised targets (pick a strategy)

- Strategy A (two-length-family system):
 - Make narrow attachment the “precision/cutting” family:
 - Ideal overall length: **190–205 mm** (Benchmark-backed to cutting comb lengths) ²
 - Acceptable: **185–210 mm**
 - Too long: **>215 mm** for “fast cutting comb” role (Inference)
- Strategy B (single-length-family system for modular consistency):
 - Keep narrow attachment near wide attachment:
 - Ideal: **220–230 mm** (Benchmark-backed by long cutting comb) ²⁴
 - This improves system consistency but sacrifices “short comb speed.” (Engineering inference)
- **Tooth geometry (prototype-validation driven)**
 - Fine tooth pitch: **0.8–1.4 mm** (Inference)
 - Coarse tooth pitch (if dual zone): **1.8–3.0 mm** (Inference)
 - Tooth root thickness: **1.4–2.2 mm** (Inference)
 - Tooth tip thickness: **0.7–1.2 mm**, rounded (Inference)

Confidence level: Medium on length decision (depends on positioning); low-medium on tooth parameters pending tests.

Master sizing table

The table below is intentionally “CAD-briefable”: ideal target, acceptable range, and stop thresholds. Items not present in the chat prompt are labeled “**not found in chats**”; values provided are a mix of benchmark-backed and inference, with basis indicated.

Basis codes: **BB** benchmark-backed, **EI** engineering inference, **PV** requires prototype validation.

Part	Current size (chat)	Recommended size (ideal)	Acceptable range	Too small	Too large	Notes to CAD designer
Main comb body	208.01 mm L	210 mm L (EI/ BB)	200–220 mm (BB)	<190 mm	>230 mm	Keep near 200– 220 unless you want “short cutting” family. Radii: 25–40% wall thickness; min abs ~0.5 mm. ²⁶
Round handle connector	242.65 mm L; tip ~3.11 mm	230 mm L; tip 2.3 mm (EI/PV)	215–242 mm; tip 1.8– 3.0 mm (BB/ PV)	<210 mm	>245 mm	If this is a parting-tail component, tip likely wants slimming + radius for comfort. ²⁷
Flat handle connector	unspecified	228 mm L; grip 22×10 mm (EI)	215–242 mm; grip width 18–26 mm, thickness 8– 12 mm (EI/ BB)	<210 mm	>245 mm	Use oval/flat to index orientation and reduce pinch hotspots (radii strategy). ²¹
Double handle connector	242.8 mm L; grip width 27.2 mm	235 mm L; grip major 28–30 mm (EI)	230–240 mm; major 26–32 mm (EI)	<220 mm	>245 mm	Width is promising; manage hair snag zones and wobble amplification at interface. ²⁸
Wide comb attachment	223.23 mm L	230 mm L (BB/ EI)	220–240 mm (BB)	<215 mm	>245 mm	Tooth geometry must be prototype- validated. Prioritize snag- free tooth bases and generous root fillets. ²⁹

Part	Current size (chat)	Recommended size (ideal)	Acceptable range	Too small	Too large	Notes to CAD designer
Narrow comb attachment	223.67 mm L	Option A: 198 mm L (BB) or Option B: 225 mm L (BB)	A: 185–210 mm; B: 220–230 mm	A: <180; B: <215	A: >215; B: >240	Decide whether you want a true cutting-length module (faster) or consistent-length system (simpler muscle memory). ³⁰

Weight and all-day ergonomics table

Published weights for combs are inconsistently reported across retailers; where credible weights exist, tail combs can be extremely light (single-digit grams). ³¹ Because TAYLKOMB is modular (more parts, connectors), you should target “light but controlled,” using weight only as a secondary dial—aligned with ergonomic guidance that weight and balance affect dexterity and fatigue. ¹⁰

Key biomechanical anchors:

- Precision grip handle diameters **8–16 mm** (nominal ~12 mm). ⁵
- Power grip handle diameters **30–50 mm** (often too big for a comb unless you provide a palm-support form). ⁵
- Tool weight should not put the center of gravity far from the handle; keep the tool feeling “easy” in use orientation. ³²

Recommended targets (engineering inference; validate with prototypes):

Configuration	Target piece weight	Target assembled weight	Balance recommendation	Grip logic	Fatigue risk notes
Main comb body alone	10–20 g (EI)	n/a	Center of mass near center of grip zone	Keep spine light; stiffness via geometry not mass	Low fatigue if <20g
Main + round connector	20–40 g (EI)	25–50 g (EI)	Balance point 10–25 mm forward of pinch point for precision parting	Effective grip Ø 10–14 mm	Risk rises if tip-heavy
Main + flat connector	20–45 g (EI)	30–55 g (EI)	Slightly forward bias OK; don't exceed “wrist flick” fatigue	Oval 18–26 by 8–12 mm	Watch pinch hotspots

Configuration	Target piece weight	Target assembled weight	Balance recommendation	Grip logic	Fatigue risk notes
Main + double connector	30–60 g (EI)	40–70 g (EI)	Keep balance within the palm support region; avoid “lever-heavy” feel	Major axis 26–32 mm if you want lower grip force	Moderate fatigue if >70 g
Wide attachment use	10–25 g (EI)	+5–15 g over baseline	For detangling strokes, neutral balance (near grip center) reduces wrist load	Provide slip-resistant microtexture (not rubbery tack)	Avoid head-heavy wide-tooth module
Narrow attachment use	8–20 g (EI)	keep as light as possible	For cutting, bias toward neutral/slight rear to reduce tip drift	Precision-friendly profile	Overweight reduces speed and control

Lock vs validate:

- Lock: handle “effective diameter” targets (8–16 mm precision; scaled oval alternatives) as structural requirements. ⁵
- Validate: exact grams and balance point with **simple adjustable ballast prototypes** (tape-on weights) before locking production mass.

Modular connector recommendation

Recommendation: keyed rail/polygon + detent retention + seam shielding (hybrid)

Why this is the best-fit for a 10,000+ swap salon environment (engineering inference):

- **Anti-wobble must be geometric**, not friction-based—hair oils, disinfectants, and wear will defeat interference-only strategies over time.
- A **keyed interface** (e.g., double-D, hex, or shallow dovetail rail) prevents rotation and allows a long bearing surface for stability.
- Retention should be via a **detent** (spring feature or molded snap nub) located away from hair contact surfaces, behind a seam shield, to reduce snagging and debris ingress.

How to implement in CAD (dimension targets)

These targets are inference, intended as CAD starting points; validate via prototypes + tolerance stack.

- **Engagement length (bearing length): 18–28 mm**
- Too short: <15 mm → wobble risk increases
- Too long: >32 mm → harder insertion, more debris trapping, higher tolerance sensitivity
- **Key depth (anti-rotation flat / spline / rail): 0.8–1.6 mm** effective depth (per side)

- **Lead-in chamfer: 0.8–1.5 mm** at 30–45° (hair-shedding + easy insertion)
- **Seam strategy:** step seam inside a **recessed “moat”** with a **hair-shed chamfer** so hair glides past rather than catching.

If you use any snap-fit feature

Follow snap-fit design best practices:

- Prefer **tapered cantilever arms** (diminishing cross section) to achieve more uniform strain than constant-section arms. ³³
- Design permissible strain conservatively (e.g., amorphous ~70% of yield strain; GF materials without yield ~½ elongation at break)—this is especially important for repeated cycling. ³⁴
- Put snap features away from hair paths and design for **burr-free** surfaces. This aligns with premium comb expectations that stress snag-free geometry. ³⁵

Tolerance logic (production realism)

- Injection-molded interfaces require tolerance realism: tooling accuracy and resin shrink/warpage mean you should expect dimensional variation that scales with feature size and material choice. ⁹
- Use published tolerance frameworks (e.g., ISO tolerance standards for molded parts exist) as the backbone of your tolerance plan, then tighten only critical features. ³⁶

Material recommendation

Benchmark signals from premium brands

- Y.S. Park ³⁷ tail comb (YS-101) is listed at **216 mm** and highlights heat resistance and specialized tooth pitch; it also includes a disinfection guidance note (“soap and water only”), which is important for your chemical-resistance goals. ³⁸
- Sam Villa ³⁹ combs state **computerized molds eliminate parting lines**, use silicone/graphite resins, and claim heat resistance up to **450°** (market expectation: snag-free, heat tolerant, durable). ¹¹
- Hercules Sägemann ⁴⁰ emphasizes **handmade combs** and that premium combs are made from **100% vulcanised natural rubber**; sellers position these as heat/chemical resistant and burr-free. ⁴¹

Implications for TAYLKOMB materials (engineering inference)

Because your explicit requirement includes tolerance to **bleach, disinfectants, oils**, and heavy salon use, you likely need **engineering polymers** with strong chemical resistance and fatigue performance at thin sections.

A production-practical material strategy:

1) **Comb teeth + high-wear interfaces** (need stiffness + fatigue + low friction):

- Primary candidate: **POM (acetal)** for stiffness/low friction and tight molding capability (commonly used in precision components). (Inference; supported by general manufacturing materials guidance that POM is used for tight tolerances and stability) ⁴²
- Alternative candidates for heat/chemical: selected grades of **PBT/PPS** (higher cost), depending on required disinfectant exposures and moldability of thin teeth. (Inference)

2) **Handle connectors (comfort + toughness):**

- Candidate: **PC/ABS class** or similar tough resin for impact resistance and premium feel, with microtexture rather than soft overmold to preserve cleanability. (Inference; DFM radii/draft guidance applies) ⁴³

3) **Premium “feel” option** (if you want Hercules-like identity):

- Explore **hard rubber / ebonite-like** formulation for a premium line, but you must validate manufacturability and tight tolerance locking surfaces. (Inference; benchmark indicates vulcanised rubber premium combs exist) ⁴¹

Chemical disinfection reality check

- Barbicide ⁴⁴ describes typical immersion disinfectant contact times (often **~10 minutes**) and emphasizes that only hard, non-porous items can be disinfected (e.g., glass/metal/plastic/synthetics). ⁴⁵
- If your target salons rely on immersion disinfectants, your materials and finishes should tolerate repeated cycles—this is an explicit “prototype validation” item (exposure testing) because chemical compatibility is **grade-specific** and depends on concentration/time/temperature. ⁴⁶

CAD revision plan and prototype validation plan

Change now (Rev B actions with high expected ROI)

- 1) **Define two length families** (or commit to one), and normalize modules accordingly (see master table). ⁴⁷
- 2) **Redesign all hand-contact edges** to comply with radii guidance: radii **25–40% of wall thickness**, **≥0.5 mm** in demanding areas; add additional “comfort radii” where fingers pinch. ⁶
- 3) **Lock grip sizing envelopes**: precision 8–16 mm effective diameter; design oval/flat sections using these equivalences. ⁵
- 4) **Modular seam shielding**: ensure no seam lip or step in hair path; add hair-shed chamfers and flush transitions informed by “no parting lines/snags” premium expectations. ¹¹
- 5) **Connector anti-wobble**: add keyed bearing surfaces + longer engagement length (18–28 mm) and explicit anti-rotation geometry. ²²
- 6) If using snap features: **taper cantilever arms** and design within permissible strain guidance. ⁷

Keep for now (tentatively acceptable pending validation)

- Main comb body overall length near **~208–210 mm** looks reasonable; keep unless you reposition your “family strategy.” ⁸
- Wide attachment length near **223 mm** is close to a proven long-comb benchmark (~222 mm). ⁴⁸

Do not lock yet (highest uncertainty dimensions)

- **Tooth pitch, tooth length, tooth thickness (root and tip)** for both attachments. (Not found in chats; must be tuned by hair-type trials.) ³⁵
- **Round connector tip diameter** and taper style (precision vs comfort vs strength).
- **Exact engagement tolerances** for the modular interface until injection tolerance strategy and material are chosen. ⁹
- **Disinfection chemical compatibility** and surface finish response (clouding, embrittlement, odor retention). ⁴⁹

Top geometry revisions (CAD checklist)

- 1) Standardize radii strategy across system (comfort + structural) per Covestro guidance. ⁶
- 2) Add lead-in chamfers and hair-shedding transitions at every seam.
- 3) Ensure tooth bases have generous fillets; remove sharp re-entrant corners. ⁶
- 4) Avoid constant cross-section cantilever snaps; taper them if used. ³³
- 5) Place parting lines away from hair contact zones where feasible; chase “seamless glide.” ¹¹
- 6) Add tactile indexing to round sections (micro-flat or subtle ridge) for orientation.
- 7) Ensure grip zones avoid pressure points; provide microtexture (cleanable) instead of sticky elastomer. (Inference)
- 8) Design interface to be debris tolerant: avoid deep hair-trapping grooves near latch.
- 9) Use ribs instead of thick sections for stiffness (rib thickness ~40–60% of wall thickness is a common injection guideline). ⁵⁰
- 10) Ensure draft on molded parts consistent with molding guidance (add in CAD now to avoid later geometry churn). ⁵¹

Prototype validation plan (exact tests to run before locking)

- 1) **Comfort test (static + dynamic):** 30 min continuous use per connector; record hot spots and pinch fatigue.
- 2) **Fatigue test:** 2-hour repetitive sectioning + detangling simulation; compare handle variants.
- 3) **Repeated swap test:** 10,000 cycle bench test (manual jig), inspect wear, insertion force drift, wobble growth.
- 4) **Drop test:** 1.0 m onto hard surface, multiple orientations; inspect tooth cracks and interface failure.
- 5) **Snag test:** comb through wet curly hair + extension hair bundle; look for seam catches and tooth-base snags.
- 6) **Heat exposure:** blow-dryer + iron proximity tests aligned with market expectations (premium combs advertise high heat resistance). ⁵²
- 7) **Chemical exposure:** follow salon disinfectant contact time (e.g., ~10 minutes immersion typical) + rinse/dry cycles; check embrittlement/warp/surface changes. ⁴⁵
- 8) **Breakage test:** torsion and bending at tail tip and tooth zone.
- 9) **Working stylist trial:** 3–5 stylists, 1–2 full workdays, structured survey + observed handling.

Optional parametric appendix

This appendix is a CAD-support scaffold. It uses relationships rather than rigid numbers, so you can drive a parametric model and enforce consistency across all modules.

Core parameters (define once)

Let:

- **L_total** = overall length of module (mm)
- **L_teeth_zone** = length of tooth-bearing region along spine (mm)
- **L_handle_zone** = grip/handle region length (mm)
- **L_interface** = connector engagement/bearing length (mm)
- **t_wall** = nominal structural wall thickness (mm)
- **R_feat** = general feature radius (mm)
- **p_tooth** = tooth pitch (mm)
- **t_root** = tooth root thickness (mm)
- **t_tip** = tooth tip thickness (mm)

Suggested relationships (engineering inference, consistent with DFM guidance):

- **L_total = L_teeth_zone + L_handle_zone**
- **L_interface = 0.08–0.12 × L_total** (typical starting band; e.g., 18–28 mm for ~230 mm modules)
- **R_feat ≈ 0.25–0.40 × t_wall**, and **R_feat ≥ 0.5 mm** in demanding structural areas. ⁶
- If a cantilever snap is used: taper the arm thickness so the free end thickness is materially reduced vs root (avoid constant cross section). ³³

Tooth array rules (starting templates; validate in prototypes)

- Narrow comb:
 - **p_tooth_fine** = 0.8–1.4 mm; **t_root** = 1.4–2.2 mm; **t_tip** = 0.7–1.2 mm
- Wide comb:
 - **p_tooth_wide** = 4–8 mm; **t_root** = 2.2–3.2 mm; **t_tip** = 1.2–1.8 mm

Datum logic for modular interface

- Establish a single **Interface Datum A** on the mating face plane.
- Establish **Datum B** as the anti-rotation key plane/axis.
- Control wobble by constraining:
 - one **axial stop** (Datum A contact)
 - one **radial bearing cylinder/rail** (controls lateral play)
 - one **key feature** (controls rotation)

Notes on source constraints

Your provided prompt contains draft dimensions but does not include files, CAD screenshots, detailed cross-sections, tooth counts, tooth geometry, engagement geometry, or existing interface architecture. Those

specifics are therefore treated as “**not found in chats**” and are addressed via benchmark-backed ranges and engineering inference, with explicit calls for prototype validation where uncertainty is intrinsically high.

1 2 15 30 47 **YS Park 339 Comb Even Fine Teeth**

https://ysparkusa.com/products/ys-park-comb-even-fine-teeth-339?srsId=AfmBOoqkHgYTiR6T_pEbpa9E_bFvRSJ5RNWAB5eSNcggEsL39hCgWBz&utm_source=chatgpt.com

3 13 18 19 27 38 **YS Park 101Tail Comb - 8.5"**

https://ysparkusa.com/products/ys-park-tail-comb-101?srsId=AfmBOOpQt00RIOfBund6IsXr-KNvEsRZRtu3Q3Tcl8Ky9VN3QMQ8KPWe&utm_source=chatgpt.com

4 23 **Hercules Sagemann | Combing Ebonite**

<https://www.hairandbeautyonline.com/en/brand/hercules-sagemann/>

5 20 21 28 **CCOHS: Hand Tool Ergonomics - Tool Design**

https://www.ccohs.ca/oshanswers/ergonomics/handtools/toolDesign.html?utm_source=chatgpt.com

6 26 43 **solutions.covestro.com**

<https://solutions.covestro.com/-/media/covestro/solution-center/story/brochures/strategic-radii-in-structural-part-design.pdf?hash=3B2CD508B82ED05C15F5020BE55D83CD&rev=afedb8d7807f4911a2f97b78f4f1e025>

7 12 33 34 **solutions.covestro.com**

<https://solutions.covestro.com/-/media/covestro/solution-center/brands/downloads/imported/1556891135.pdf>

8 11 24 25 29 35 48 **Signature Series Long Cutting Comb | Sam Villa Combs**

https://www.samvilla.com/products/long-cutting-comb?srsId=AfmBOOpUR-EV7pldoVV_jWR7rjKMgjm3uNs9LNweIQNCZ6zgWWPoMpBC

9 22 **Understanding Injection Molding Tolerances**

https://www.protolabs.com/resources/blog/injection-molding-tolerances/?utm_source=chatgpt.com

10 **ERGONOMIC GRIP DESIGN CONSIDERATIONS**

https://www.avient.com/sites/default/files/2021-11/avient-design-ergonomic-design-guide.pdf?utm_source=chatgpt.com

14 16 17 39 **Thin Wall Injection Molding | Materials and Design Tips**

https://www.protolabs.com/resources/blog/thin-wall-injection-molding/?utm_source=chatgpt.com

31 44 **Y.S.PARK 101 TAIL COMB**

https://www.fineedgesupply.com/products/y-s-park-101-tail-comb?srsId=AfmBOOrJeWJS1ubTcGVD-gax6-220txNqkvTlpMKUu5eHmX_PvpRSQXr&utm_source=chatgpt.com

32 **HAND TOOL DESIGN**

https://ergo.human.cornell.edu/studentdownloads/DEA3250pdfs/Hand%20Tools.pdf?utm_source=chatgpt.com

36 37 **ISO 20457:2018 - Plastics moulded parts — Tolerances ...**

https://www.iso.org/standard/68097.html?utm_source=chatgpt.com

40 41 **Hercules Sägemann EN – Premium Combs and Brushes**

<https://www.hercules-saegemann.de/en/>

42 **Create custom parts with injection molding**

https://www.hubs.com/injection-molding/plastic/?utm_source=chatgpt.com

45 46 49 **BARBICIDE® Back to Work Plan**

https://www.barbicide.com/back-to-work-plan/?utm_source=chatgpt.com

50 **Injection Molding Basics: An Intro to Designing Plastic Parts**

https://www.protolabs.com/resources/design-tips/injection-molding-basics/?utm_source=chatgpt.com

51 **Draft Angle Guidelines for Injection Molding**

https://www.protolabs.com/resources/design-tips/improving-part-moldability-with-draft/?utm_source=chatgpt.com

52 **Signature Series Wide Cutting Comb | Sam Villa Combs**

https://www.samvilla.com/products/wide-cutting-comb?srsItid=AfmBOorWdy1R2ZupLjqNTTsoggE_ssEA72rIv_AFKnChiSDYuWR-CIM8